

Course “Software Verification and Certification”

Course Code: CS 448/648

Term: *Spring 2026*

Lecture#1: 6 Jan 2026

Dr. Gourinath Banda

Professor and Lead_PI - ACPS Lab

Department of Computer Science and Engineering

Indian Institute of Technology Indore

Venue @ L-01, LHC

Course Introduction

- Term “Spring 2026”
- This course is an outcome of the project “Enabling Inclusive, Energy-Aware, and Formally Correct Computing: A Holistic Approach to Responsible System Development”

Thanks to “Mozilla Foundation, California, USA” for sponsoring this course design



Course Introduction [1/3]

- Term “Spring 2026”
- This course is an outcome of the project “Enabling Inclusive, Energy-Aware, and Formally Correct Computing: A Holistic Approach to Responsible System Development”

Thanks to “Mozilla Foundation, California, USA” for sponsoring this course design



Course Introduction [2/3]

The agreement that Free and Open Source Software (FOSS) typically presents during installation, which includes clauses that disclaim any legal guarantees, is an **End-User License Agreement (EULA)** or simply a **Software License Agreement**. These agreements often contain a specific **warranty disclaimer** and a **limitation of liability** clause to protect the developers. 📄

Key Clauses in FOSS Agreements

While the entire document is the license agreement, the specific provisions that address the lack of legal guarantees are: 📄

- **Warranty Disclaimer:** This clause explicitly states that the software is provided "**AS IS**" and without any warranty of any kind, either expressed or implied. This means the authors make no claims that the software will perform in any given way, be error-free, or even function correctly for a particular purpose (e.g., merchantability or fitness for a particular purpose).
- **Limitation of Liability:** This clause further ensures that the authors or copyright holders are not liable for any damages (e.g., lost profits, data loss, business interruption) arising from the use or inability to use the software, even if they have been advised of the possibility of such damages. 📄

Purpose

These clauses are essential for FOSS developers because the software is often provided free of monetary charge and developed by a wide range of contributors. By including these disclaimers, developers protect themselves from extensive legal and financial exposure if the software causes problems for the user. 📄

Agreeing to the terms (often via a "clickwrap" agreement during installation) is a user's acknowledgment of these inherent risks and the conditions under which they are granted permission to use the software. 📄

Opening Context ... (this course is a response to a global inflection point)

- Why this course, why now?
 - *Ofcourse ... blah ... blah ... blah*
 - ***“Software today does not merely compute ---
it decides, governs, and controls.”***
 - This course “Software Verification and Certification”
 - sits at the intersection of
 - computer science,
 - ethics, safety, and
 - public trust.
- PLEASE NOTE **“This is not a routine elective”**

Motivation [1/2]

- Modern software systems now:
 - “controls: vehicles, medical devices, drones, energy grids”
 - “mediate: financial transactions and public services”
 - “shape: speech, access, and opportunity through **algorithms**”
- Such software IF fails:
 - the cost is beyond technicalities
 - Consequences are: human/legal/societal.
 - This course is a step towards:
 1. Responsible computing
 2. Integrity in Software
 3. Towards reliable technologies impacting people at large in all walks of life

Motivation [2/2]

- *Shift is From:*
 - *Does It Work?"*
to
 - *"Can We Trust It?"*
- Traditionally, focus was on:
 - Does the program compile?
 - Does it pass test cases?

Deeper questions on systems' correctness...

- Questions such as:
 - Is the software correct for **all possible executions**?
 - Does it guarantee **safety**, not just average behavior?
 - Can we prove **critical properties** rather than hope for them?
 - Can an open-source system be **certified** and **trusted at scale**?

Key difference: Verification Vs. Testing

- Testing answers
 - “Did the program work for these inputs?”
- Verification asks and is the answer to:
 - ““Can the program ever violate this property?””
- Differences between:
 - Formal Analysis (aka **mathematical**) Vs. Dynamic Analysis (aka **testing**)
 - Empirical confidence vs Mathematical assurance
 - Manual reasoning Vs. Automated reasoning about verification of systems

Verification Vs. Testing: understanding the difference

- You will learn why:
 - Testing can demonstrate presence of bugs, but never their absence.
 - Formal verification can prove properties over infinite behaviors.
 - Certification requires evidence, not intuition.

This distinction is central to **regulatory acceptance**, which is why verification **underpins** certification.

Module: Formal Verification (The Core methods and techniques)

- Algorithmic Verification comprises:
 - Model checking
 - State-space exploration
 - Temporal logic (safety & liveness)
- Deductive Verification comprises:
 - Logical proofs
 - Pre/post-conditions, invariants
 - Theorem proving
- Hybrid Approaches comprising
 - Combining static analysis with model checking
 - Scalable industrial verification pipelines

An inspiration towards Sustainability and Technology...

- Gourinath Banda: Visual Profile

“An inspiring interaction with President and B.R. APJ Kalam about a key tech for space missions”



SDV Team

Module: Formal Verification (The Core methods and techniques)

- Algorithmic Verification
 - • Model checking • State-space exploration • Temporal logic (safety & liveness)
- Deductive Verification
 - • Logical proofs • Preconditions, postconditions, invariants • Theorem proving
- Hybrid Approaches
 -

Module: Verification Languages, Tools, and Standards

- Verification is not just theory, it is embedded in its
 - Specification languages
 - Modeling formalisms
 - Toolchains
 - Coding standards
- **You will learn:**
 - Why languages matter for verifiability
 - How coding standards influence analyzability
 - How verification tools integrate into real development workflows

Module: Verification Languages, Tools, and Standards

- Verification is not just theory, it is embedded in its
 - Specification languages
 - Modeling formalisms
 - Toolchains
 - Coding standards
- **You will learn:**
 - Why languages matter for verifiability
 - How coding standards influence analyzability
 - How verification tools integrate into real development workflows

This sets the stage for **certification**, where compliance must be **repeatable and auditable**.

Module: **Open Source Vs. Closed -- Certification**

- Tension between:
 - Openness vs Verifiability
 - Rapid innovation vs Rigorous assurance
- **You will indulge in:**
 - Industrial case studies
 - Certification of open-source components
 - Responsibility specification and analysis

Course Outcomes and Student Pathways

- you will be able to:
 - Apply formal verification to ensure safety, liveness, and correctness
 - Embed correctness checks across the software lifecycle
 - Reason about correctness, -verification, certification and responsibility, not just functionality

Student Pathways

- This Course is for “Everybody who is into critical software systems development”
 - Industry-oriented students → safety-critical systems, compliance roles
 - Policy-aware technologists → responsible AI, digital governance
 - Open-source contributors → verifiable, trustworthy software
 - Entrepreneurs → huge untapped market is waiting!
 - Military officials → Are we procuring trustworthy systems?
 - Software quality personnel → What are the tools you can indulge in?

Course Outcomes and Student Pathways

- you will be able to:
 - Apply formal verification to ensure safety, liveness, and correctness
 - Embed correctness checks across the software lifecycle
 - Reason about correctness, -verification, certification and responsibility, not just functionality
 - Openness vs Verifiability
 - Rapid innovation vs Rigorous assurance

An inspiration towards Sustainability and Technology...

- Gourinath Banda: Visual Profile

“An inspiring interaction with President and B.R. APJ Kalam about a key tech for space missions”



SDV Team

Our Coordinates...

<https://maps.app.goo.gl/YVjn2wcxBTWZ7geo9>



Lat, Long: 22.529335, 75.923259



Introduction

- **I believe:**
 - **State of the Art in Innovation is that state**
 - **When Sustainability drives** modern technological advancements
 - My contributions focus on IoT, Software-Defined Vehicles (SDVs), Responsible Computing, and Digital Payments.
 - Recognized at **national and international levels** for research and practical innovations
 - This presentation summarizes my **achievements, patents, and entrepreneurial leadership in sustainability-focused technology.**

Background & Strengths

- Modern civilisations are all about automation
 - **Often realised through IoT, Cyber Physical Systems, AI/ML**
- My Expertise and Skillsets are all in:
 - **Cyber-Physical Systems, IoT, and Sustainable Computing.**
 - **Cutting edge Academic & Industry Contributions through projects leading to real-world impact**
- My Core Strengths:
 - Developing scalable and sustainable technology solutions.
 - Bridging the **gap between academia and industry**
 - **Driving national and international innovation efforts in sustainability and technology**

My Entrepreneurial Vision

- Founded two startups:
 1. Vedblockchain Systems Pvt. Ltd. + Recognized globally for sustainability and digital transformation.
 2. DigiTraWS Pvt. Ltd. + Focused on accountability-driven digital transformation towards sustainable governance and enterprising systems.
- Major recognitions:
 1. Vedblockchain recognized by UAE Prime Minister's Office and awarded Gold at mGov 2023. (Presented by Egypt's President)
 - Details on the next slide
 2. DigiTraWS received ₹12 Lakhs for its sustainability-focused innovations.
 - Details on the next slide

Dr. Gourinath's Product wins "Best-mGov" Gold Award

- Egyptian President gave Gold Award to our Sustainability-scoped Timely Project "BlockBill"
 - Prestigious Inter-governmental WGS Summit 2023
 - Accepted in the prestigious Industry IoT Consortium journal of IBM/OMG USA
- Product "BlockBill"
 - A Paperless Billing Solution
 - Eliminates paper receipts
 - Reduces environmental emissions/waste
 - Ensures secure digital transactions with full transparency
 - Recognized globally for financial sustainability and digital efficiency.
 - Proof/News links: IITNews YouTubeLink TwitterLink PTILink [InstagramLink](#)



Dr. Gourinath's Browser wins Ideation and Prototype Awards from MEITY+CDAC

- IWBD C Challenge winner
 - Awarded by MEITY + CDAC
 - Winner of Ideation Phase
 - Winner of Prototype Phase
 - For: **India's indigenous, secure, energy-efficient web browser.**
 - Sustainability Impact:
 - Low computational overhead, ensuring optimized power consumption.
 - Privacy-centric, reducing data wastage and enhancing security.
 - Promotes local innovation, reducing reliance on foreign technologies
- NEWS Proof: [IWBD C MEITY Results Page](#)

Dr. Gourinath bestowed with Responsible Computing Challenge

- Awarded RCC India worth US\$25000 = INR 20 Lakhs
- One of the 6 awardees recognised by USAID and Mozilla Foundation USA
 - Thrust Areas: Focus on Sustainability Ethical AI and low-energy computing frameworks.
 - Responsible, sustainable AI development.
 - Enabling low-resource digital transformation globally.
 - Successfully made UG education interventions
 - NEWS/Proof: [Mozilla Corp. Link](#)



USAID
FROM THE AMERICAN PEOPLE



OneloT Tech Stack: Only Sustainable IoT Solution ever

- Core Contributions:
 - A new manifest based IoT Approach: First ever!
 - A new language developed for sustainable IoT solution
 - Energy-efficient communication protocols
 - Smart networks optimization
 - Industrial IoT adoption for smart cities.
 - [Article Link](#)

